

UNIT I

Introduction to International Economics and Scope of International Economics

Introduction to International Economics

International economics is the branch of economics that studies economic relations between different countries. It explains how nations exchange goods services capital technology and resources across national borders. The subject focuses on international trade international finance and global economic cooperation. It helps to understand how countries interact economically and how these interactions influence production consumption employment and economic development.

International economics studies the principles that guide trade between nations. Countries differ in natural resources labour capital technology and climate. Because of these differences countries cannot produce all goods efficiently. Therefore they engage in international trade to obtain goods that they cannot produce easily or cheaply. International economics explains why trade takes place and how countries benefit from it.

Another important aspect of international economics is international finance. It studies foreign exchange markets balance of payments international investment and monetary relations between countries. It also examines the impact of globalization trade policies tariffs quotas and trade agreements on national economies.

International economics plays an important role in economic development. Through international trade countries can expand their markets increase production improve technology and raise living standards. It also helps policymakers design trade policies that promote growth and economic stability.

Scope of International Economics

The scope of international economics is very wide because it deals with different aspects of economic relations between nations. The major areas included in the scope of international economics are as follows.

International Trade

International trade is the most important part of international economics. It studies the exchange of goods and services between countries. It explains why countries trade what goods they trade and how trade benefits participating nations. International trade also examines trade policies such as tariffs quotas and trade restrictions.

Theory of International Trade

This area studies the theoretical explanations of international trade. Economists developed different theories such as absolute advantage comparative advantage and factor endowment theory to explain the basis of trade between countries. These theories explain specialization production efficiency and gains from trade.

Commercial Policy

Commercial policy refers to government policies that influence international trade. It includes tariffs import restrictions export promotion subsidies and trade agreements. Governments use these policies to protect domestic industries encourage exports and regulate imports.

Balance of Payments

Balance of payments is the systematic record of all economic transactions between a country and the rest of the world. It includes exports imports capital movements and financial transfers. International economics studies the causes of balance of payments disequilibrium and methods to correct it.

Foreign Exchange

Foreign exchange deals with the exchange of one currency for another. International economics studies foreign exchange markets exchange rate determination and the effects of currency fluctuations on international trade and investment.

International Financial Institutions

International economics also studies the role of international financial institutions such as International Monetary Fund World Bank and World Trade Organization. These organizations promote international cooperation financial stability and global trade.

International Capital Movements

International capital movements refer to the flow of capital between countries in the form of foreign investment loans and financial aid. These movements influence economic development employment and industrial growth in different countries.

Economic Integration

Economic integration refers to cooperation between countries to reduce trade barriers and promote economic unity. Examples include free trade areas customs unions common markets and economic unions. Integration helps increase trade investment and economic growth.

Conclusion

International economics is an important field of study that explains economic relations between countries. It helps to understand international trade financial relations and economic cooperation among nations. The subject plays a vital role in promoting economic development efficient resource allocation and improved living standards across the world.

Gains from International Trade

International trade refers to the exchange of goods and services between different countries. It plays an important role in the economic development of nations. Countries engage in international trade because they cannot produce all goods efficiently due to differences in resources technology climate and skills. Through international trade countries can specialize in producing goods in which they are efficient and obtain other goods from foreign countries. This leads to many benefits which are known as gains from international trade.

Efficient use of resources

International trade helps countries use their natural and human resources efficiently. Every country has different resources such as labour capital land and technology. By specializing in the production of goods that use their abundant resources countries can achieve higher productivity and reduce wastage of resources.

Specialization in production

One of the major gains from international trade is specialization. Countries specialize in producing goods in which they have cost advantage or higher efficiency. Specialization increases output and improves the quality of goods. When countries specialize and trade they can produce more goods than they could produce individually.

Increase in world production

International trade increases the total production of goods and services in the world. When each country specializes in the production of goods in which it has advantage global production increases. This leads to better availability of goods and services for consumers.

Expansion of markets

International trade allows producers to sell their goods not only in domestic markets but also in foreign markets. This expansion of markets encourages large scale production. Large scale production reduces cost of production and increases efficiency.

Variety of goods

Through international trade consumers get access to a wide variety of goods and services from different countries. Some goods cannot be produced in certain countries due to climatic or geographical conditions. Trade allows people to enjoy products from different parts of the world.

Lower prices

International trade increases competition among producers from different countries. This competition leads to improvement in quality and reduction in prices. Consumers benefit because they can purchase goods at lower prices.

Improvement in technology

International trade promotes the transfer of technology and knowledge between countries. Countries learn new production techniques and management methods from other nations. This improves productivity and industrial development.

Increase in employment opportunities

Expansion of industries and production for export creates more employment opportunities. Export oriented industries require more labour and resources which increases job opportunities and income levels.

Economic growth and development

International trade contributes to economic growth by increasing national income production and investment. Countries earn foreign exchange through exports which can be used to import capital goods machinery and technology for development.

Better international relations

Trade creates economic cooperation between countries. Countries that trade with each other develop friendly economic and political relations. International trade promotes peace cooperation and mutual understanding among nations.

Conclusion

International trade provides many benefits to countries such as efficient use of resources specialization expansion of markets and improvement in technology. It increases production employment and economic growth. Therefore international trade is an important factor for the development and prosperity of nations.

Classical Theory of International Trade and Mercantilism

Classical Theory of International Trade

The classical theory of international trade explains the reasons why countries trade with each other and how they gain from trade. The theory was developed by classical economists such as Adam Smith and David Ricardo. According to the classical theory international trade occurs because countries differ in their ability to produce goods efficiently. These differences in production efficiency create opportunities for specialization and trade.

The classical theory states that each country should specialize in producing goods that it can produce at lower cost compared to other countries. When countries specialize and exchange goods with each other total production increases and both countries gain from trade.

The classical theory mainly focuses on the role of labour productivity and cost differences in determining international trade. It assumes that labour is the main factor of production and that production costs depend on labour efficiency.

The theory also emphasizes the importance of free trade. Classical economists believed that trade should not be restricted by government policies because free trade allows countries to specialize according to their advantages and increase global production.

Another important idea in classical theory is the concept of division of labour. When countries specialize in producing particular goods they become more efficient which leads to higher output and economic growth.

The classical theory laid the foundation for modern international trade theories. It explains that international trade benefits all participating countries by increasing production improving efficiency and providing access to a variety of goods.

Mercantilism

Mercantilism was an early economic doctrine that influenced European trade policies during the sixteenth seventeenth and early eighteenth centuries. According to mercantilist thinkers the wealth and power of a nation depended on the accumulation of precious metals such as gold and silver.

Mercantilists believed that a country should export more goods than it imports in order to achieve a favourable balance of trade. A favourable balance of trade means that the value of exports is greater than the value of imports. When exports are greater than imports a country receives gold and silver from other nations which increases national wealth.

To achieve this objective governments adopted several policies to control trade. They encouraged exports by providing incentives and discouraged imports by imposing high tariffs and restrictions. Governments also promoted domestic industries so that the country could produce more goods for export.

Mercantilism also supported the establishment of colonies. Colonies were considered important because they provided raw materials for domestic industries and served as markets for finished goods. This helped mercantilist countries expand their trade and increase national wealth.

Another feature of mercantilism was strong government control over economic activities. The state regulated trade production and shipping in order to maintain a favourable balance of trade.

Although mercantilism helped in the growth of trade and industry it had several limitations. It encouraged trade restrictions and ignored the benefits of free trade. Later economists such as Adam Smith criticized mercantilism and argued that international trade should be based on mutual benefits rather than accumulation of gold and silver.

Conclusion

The classical theory of international trade explains that countries gain from specialization and free trade based on differences in production efficiency. Mercantilism on the other hand focused on increasing national wealth through favourable balance of trade and accumulation of precious metals. The criticism of mercantilism led to the development of modern theories of international trade which emphasize the benefits of free trade and specialization.

Absolute Cost Advantage Trade Theory

Absolute cost advantage theory was introduced by Adam Smith. This theory explains the basis of international trade between countries. According to Adam Smith a country has absolute advantage when it can produce a commodity with fewer resources or less labour compared to another country. Countries should specialize in producing goods in which they have absolute cost advantage and import goods that other countries can produce more efficiently.

Adam Smith criticized mercantilism and supported the idea of free trade. He argued that international trade increases productivity and total world output when countries specialize according to their efficiency. Absolute cost advantage is based on the principle that if each country produces goods in which it is most efficient the total production of goods in the world will increase and both countries will gain from trade.

The theory assumes that labour is the only factor of production. It also assumes that there are two countries two commodities and labour can move freely within a country but not between countries. The theory also assumes that transportation cost is negligible and production technology is constant.

According to this theory a country should export the commodity in which it has absolute cost advantage and import the commodity in which it has absolute disadvantage. This leads to specialization efficient use of resources and mutual benefits for trading nations.

Numerical Example

Consider two countries Country A and Country B

Two commodities Cloth and Wheat

Labour hours required to produce one unit of each commodity are given below

Country A

Cloth 10 labour hours

Wheat 20 labour hours

Country B

Cloth 30 labour hours

Wheat 15 labour hours

In this example Country A requires fewer labour hours to produce cloth compared to Country B. Therefore Country A has absolute advantage in the production of cloth. Country B requires fewer labour hours to produce wheat compared to Country A. Therefore Country B has absolute advantage in the production of wheat.

According to the theory Country A should specialize in producing cloth and export cloth to Country B. Country B should specialize in producing wheat and export wheat to Country A. Through specialization both countries can increase production and gain from trade.

Before trade each country produces both goods using more resources. After trade each country concentrates on the production of goods in which it has absolute efficiency. This increases total output and both countries enjoy higher consumption.

Advantages of Absolute Cost Advantage Theory

- Encourages specialization in production
- Promotes efficient use of resources
- Increases total world production
- Improves international trade relations
- Raises living standards of people

Limitations of Absolute Cost Advantage Theory

- The theory assumes labour as the only factor of production
- It ignores transportation cost and trade barriers
- It cannot explain trade when one country has advantage in producing all goods
- It assumes constant technology and perfect competition

Conclusion

Absolute cost advantage theory explains that international trade takes place when countries specialize in producing goods that they can produce more efficiently than other countries. Through specialization and exchange countries can increase production improve efficiency and gain mutual benefits from international trade.

Comparative Cost Advantage Trade Theory

Comparative cost advantage theory was developed by David Ricardo. This theory explains the basis of international trade and shows that trade can occur even when one country has absolute advantage in producing all goods. The theory states that countries should specialize in producing goods in which they have comparative cost advantage and import goods in which they have comparative disadvantage.

Comparative advantage is based on the concept of opportunity cost. Opportunity cost means the amount of one good that must be sacrificed to produce another good. A country has comparative advantage in the production of a commodity when the opportunity cost of producing that commodity is lower compared to another country.

The theory assumes two countries two commodities and labour as the only factor of production. Labour is perfectly mobile within a country but not between countries. The theory also assumes constant technology absence of transport cost and free trade between countries.

According to comparative cost theory even if one country is more efficient in producing all goods both countries can still gain from trade. The country should specialize in producing goods where it has lower opportunity cost and import goods where its opportunity cost is higher.

Numerical Example

Assume two countries Country A and Country B
Two commodities Cloth and Wheat
Labour hours required to produce one unit of each commodity are given below

Country A
Cloth 10 labour hours
Wheat 20 labour hours

Country B
Cloth 30 labour hours
Wheat 40 labour hours

In this example Country A has absolute advantage in producing both cloth and wheat because it requires fewer labour hours than Country B. However comparative advantage depends on opportunity cost.

Opportunity cost in Country A
To produce 1 unit of cloth Country A sacrifices half unit of wheat
To produce 1 unit of wheat Country A sacrifices 2 units of cloth

Opportunity cost in Country B
To produce 1 unit of cloth Country B sacrifices three fourth unit of wheat
To produce 1 unit of wheat Country B sacrifices four third units of cloth

From this comparison Country A has lower opportunity cost in producing cloth while Country B has lower opportunity cost in producing wheat.

Therefore Country A has comparative advantage in cloth production and Country B has comparative advantage in wheat production.

According to the theory Country A should specialize in producing cloth and export cloth to Country B. Country B should specialize in producing wheat and export wheat to Country A. Through specialization and exchange both countries can gain from international trade.

Importance of Comparative Cost Theory

Explains the real basis of international trade
Shows that trade is beneficial even when one country is more efficient in producing all goods
Encourages specialization and efficient allocation of resources
Promotes expansion of international trade
Increases production and economic welfare

Limitations of Comparative Cost Theory

Labour is considered as the only factor of production
Transportation costs are ignored
The theory assumes constant technology
Trade barriers and government policies are not considered

Conclusion

Comparative cost advantage theory is an important principle of international trade. It explains that countries gain from trade by specializing in the production of goods with lower opportunity cost. Even when one country has absolute advantage in producing all goods both

countries can benefit through specialization and exchange. This theory forms the foundation for modern international trade analysis.

Heckscher Ohlin Theory and Empirical Verification

Heckscher Ohlin theory is an important theory of international trade developed by Swedish economists Eli Heckscher and Bertil Ohlin. This theory explains that international trade takes place because countries differ in their factor endowments. Factor endowment means the quantity and availability of factors of production such as labour capital land and natural resources in a country.

According to this theory a country will export goods that use its abundant factors of production intensively and import goods that require factors which are scarce in that country. The theory emphasizes that differences in factor availability are the main reason for international trade between countries.

For example a country that has abundant labour and limited capital will specialize in producing labour intensive goods. A country that has abundant capital and limited labour will specialize in producing capital intensive goods. By specializing according to factor endowment countries can produce goods more efficiently and gain from international trade.

Assumptions of Heckscher Ohlin Theory

Two countries two commodities and two factors of production labour and capital
Countries have different factor endowments
Technology is the same in both countries
Factors of production are perfectly mobile within a country but immobile between countries
There are no transportation costs and no trade restrictions
Production shows constant returns to scale
Consumer preferences are similar in both countries

Explanation of the Theory

According to the theory a labour abundant country will have lower cost of labour and therefore it will produce labour intensive goods more cheaply. Similarly a capital abundant country will have lower cost of capital and it will produce capital intensive goods at lower cost.

As a result the labour abundant country will export labour intensive goods and import capital intensive goods. On the other hand the capital abundant country will export capital intensive goods and import labour intensive goods.

This specialization and exchange increases world production and improves economic welfare of both countries.

Example

Consider two countries Country A and Country B

Country A has abundant labour and limited capital
Country B has abundant capital and limited labour

Two commodities are produced
Garments which are labour intensive
Machinery which is capital intensive

According to Heckscher Ohlin theory Country A will specialize in producing garments because it has abundant labour. Country A will export garments to Country B.

Country B will specialize in producing machinery because it has abundant capital. Country B will export machinery to Country A.

Through this specialization and trade both countries gain from international trade and resources are used more efficiently.

Empirical Verification of Heckscher Ohlin Theory

Economists tried to test the validity of the Heckscher Ohlin theory using real world data. Empirical verification means testing a theory using actual economic data. The most famous attempt to test the theory was made by economist Wassily Leontief.

Leontief studied the trade pattern of the United States which was considered a capital abundant country. According to Heckscher Ohlin theory the United States should export capital intensive goods and import labour intensive goods.

However the study showed that the United States exported labour intensive goods and imported capital intensive goods. This result was opposite to what the theory predicted. This unexpected result became known as the Leontief paradox.

Reasons for the Difference

Presence of skilled labour in the United States
Differences in technology between countries
Government trade policies and tariffs
Demand conditions in international markets
Transportation costs and trade barriers

These factors influenced the pattern of international trade and created differences between theory and actual results.

Conclusion

Heckscher Ohlin theory explains international trade on the basis of factor endowments. Countries specialize in producing goods that use their abundant resources intensively and import goods that use scarce resources. Although empirical studies such as the Leontief paradox challenged some predictions of the theory the Heckscher Ohlin model remains an important contribution to the study of international trade and economic relations between countries.

Factor Price Equalization and Income Distribution Theory

Factor price equalization theory is an important concept related to the Heckscher Ohlin theory of international trade. The theory explains how international trade affects the prices of factors of production such as wages of labour and returns on capital in different countries. According to this theory international trade leads to equalization of factor prices between countries even when factors of production do not move across borders.

Factor price means the reward received by factors of production. Labour receives wages capital receives interest land receives rent and entrepreneurship receives profit. According to the theory when countries trade goods that use different factors intensively the demand for those factors changes. As trade expands the prices of factors of production in different countries gradually move towards equality.

The theory states that a labour abundant country will export labour intensive goods while a capital abundant country will export capital intensive goods. When labour intensive goods are exported the demand for labour increases in the labour abundant country which leads to higher wages. Similarly when capital intensive goods are exported the demand for capital increases in the capital abundant country which increases the return on capital.

As trade continues wages in labour abundant countries rise and returns to capital in capital abundant countries rise. At the same time wages in capital abundant countries may fall relatively and returns to capital in labour abundant countries may decrease. In the long run the prices of labour and capital tend to become equal across countries. This process is known as factor price equalization.

Assumptions of Factor Price Equalization Theory

Two countries two commodities and two factors of production labour and capital
Countries have different factor endowments
Technology is the same in both countries
Factors of production are perfectly mobile within a country but immobile between countries
There are no transport costs and no trade barriers
Production shows constant returns to scale
Consumers in both countries have similar preferences

Income Distribution

International trade not only affects factor prices but also influences income distribution within a country. According to the theory the owners of abundant factors gain from international trade while the owners of scarce factors may lose.

In a labour abundant country labour is widely available and wages are relatively low before trade. When the country begins to export labour intensive goods the demand for labour increases which raises wages and improves the income of workers. However the return to capital may decrease because capital is relatively scarce.

In a capital abundant country capital owners benefit from trade because the country exports capital intensive goods. The demand for capital increases and the returns to capital rise. At

the same time wages of labour may increase slowly or may remain relatively lower compared to capital returns.

Thus international trade redistributes income between different factors of production within a country.

Example

Consider two countries Country A and Country B

Country A is labour abundant

Country B is capital abundant

Two goods are produced

Textiles which require more labour

Machines which require more capital

Country A specializes in producing textiles and exports textiles to Country B. As textile production increases the demand for labour rises in Country A which increases wages of workers.

Country B specializes in producing machines and exports machines to Country A. As machine production increases the demand for capital rises in Country B which increases the return to capital.

Over time wages in Country A rise and returns to capital in Country B increase. The difference in factor prices between the two countries becomes smaller which leads to factor price equalization.

Empirical Verification

Economists attempted to verify the factor price equalization theory by studying real world trade patterns and factor prices. Empirical studies examined whether international trade reduces differences in wages and returns on capital between countries.

Some studies showed partial support for the theory because trade increased demand for abundant factors and changed income distribution. However complete equalization of factor prices was not observed in reality.

Reasons for Differences between Theory and Reality

Differences in technology between countries

Transport costs and trade barriers

Government policies and trade restrictions

Differences in skill levels of labour

Limited mobility of factors and imperfect competition

Because of these reasons factor prices do not become exactly equal between countries even though trade reduces some differences.

Conclusion

Factor price equalization theory explains how international trade influences wages and returns to capital across countries. Trade increases demand for abundant factors and reduces differences in factor prices between nations. It also affects income distribution within countries by benefiting owners of abundant factors. Although complete equalization of factor prices may not occur in practice the theory remains an important explanation of the relationship between international trade and factor incomes.

Leontief Paradox

Leontief paradox is an important concept in international trade theory. It challenged the predictions of the Heckscher Ohlin theory. The paradox was discovered by economist Wassily Leontief in the year nineteen fifty three while studying the trade pattern of the United States.

According to the Heckscher Ohlin theory a country exports goods that use its abundant factors intensively and imports goods that use scarce factors. The United States was considered a capital abundant country because it had large amounts of capital and advanced technology. Therefore according to the theory the United States should export capital intensive goods and import labour intensive goods.

However the study conducted by Leontief produced an opposite result. Leontief used input output analysis to examine the structure of United States exports and imports. He compared the amount of capital and labour required to produce export goods and import substitute goods.

The results of the study showed that United States exports were more labour intensive and imports were more capital intensive. In other words the United States was exporting labour intensive goods and importing capital intensive goods. This result was completely opposite to the prediction of the Heckscher Ohlin theory.

This unexpected outcome became known as the Leontief paradox because it created a contradiction between theoretical prediction and actual trade pattern.

Explanation of the Study

Leontief examined the production requirements of United States industries. He calculated the amount of capital and labour required to produce goods worth one million dollars of exports and goods worth one million dollars of imports.

The results indicated that exports required more labour compared to imports while imports required more capital compared to exports. Since the United States was capital abundant the expectation was that exports should be capital intensive. But the actual result showed the opposite which created the paradox.

Reasons for the Leontief Paradox

Presence of highly skilled labour in the United States which increased labour productivity
Use of advanced technology which reduced the need for capital in some industries

Large demand for capital intensive goods within the domestic economy
Government trade policies and import restrictions
Availability of natural resources in other countries
Differences in labour skills and education

These factors influenced the pattern of international trade and caused the results to differ from the theoretical predictions.

Importance of Leontief Paradox

It questioned the validity of the Heckscher Ohlin theory
It encouraged economists to conduct further research in international trade
It highlighted the importance of skilled labour and technology in production
It led to the development of new trade theories and modifications to existing models

Conclusion

Leontief paradox is an important empirical finding in the study of international trade. It showed that the trade pattern of the United States did not follow the predictions of the Heckscher Ohlin theory. Although the paradox challenged the theory it helped economists understand the role of technology skills and other factors in determining international trade patterns.

Factor Intensity Reversal

Factor intensity reversal is an important concept in international trade theory related to the Heckscher Ohlin model. The concept explains a situation where the same commodity may appear as labour intensive in one country and capital intensive in another country. This happens because of differences in technology factor prices and production methods between countries.

In the Heckscher Ohlin theory it is assumed that one commodity always uses a particular factor of production more intensively than another commodity. For example one good may always be labour intensive and another good may always be capital intensive in all countries. However in reality this assumption may not always hold true.

Factor intensity reversal occurs when a commodity that uses more labour in one country may use more capital in another country. In other words the relative use of labour and capital in producing the same commodity changes across countries. This situation creates difficulties in predicting trade patterns based on factor endowment theory.

The concept becomes important because it weakens the basic prediction of the Heckscher Ohlin theory which states that countries export goods that use their abundant factors intensively.

Explanation of Factor Intensity Reversal

In some countries labour may be relatively cheap and capital may be expensive. In such a situation producers may use more labour and less capital in the production process. In another

country capital may be cheaper and labour may be expensive. Producers in that country may use more machines and less labour to produce the same commodity.

Because of these differences the same commodity may appear labour intensive in one country and capital intensive in another country. This situation is called factor intensity reversal.

When factor intensity reversal occurs it becomes difficult to determine which country will export a particular commodity based on factor abundance alone. Therefore the predictions of the Heckscher Ohlin theory become uncertain.

Example

Consider two countries Country A and Country B

Country A has abundant labour and labour is relatively cheap

Country B has abundant capital and capital is relatively cheap

Two commodities are produced Textiles and Machinery

In Country A producers may use more labour and fewer machines to produce textiles because labour is cheap. Therefore textiles appear as labour intensive goods in Country A.

In Country B producers may use advanced machines and fewer workers to produce textiles because capital is cheaper. Therefore textiles may appear as capital intensive goods in Country B.

Thus the same commodity textiles becomes labour intensive in one country and capital intensive in another country. This situation represents factor intensity reversal.

Causes of Factor Intensity Reversal

Differences in technology between countries

Differences in relative prices of labour and capital

Different production techniques used by firms

Variations in skill level of labour

Government policies and industrial structure

Importance of Factor Intensity Reversal

It shows the limitations of the Heckscher Ohlin theory

It explains why trade patterns may not always follow factor endowment predictions

It highlights the role of technology and production techniques in international trade

It encourages economists to develop more realistic trade models

Conclusion

Factor intensity reversal refers to the situation where the same commodity uses different proportions of labour and capital in different countries. Because of this the classification of goods as labour intensive or capital intensive may change across countries. This concept

shows an important limitation of the Heckscher Ohlin theory and explains why actual international trade patterns may sometimes differ from theoretical predictions.